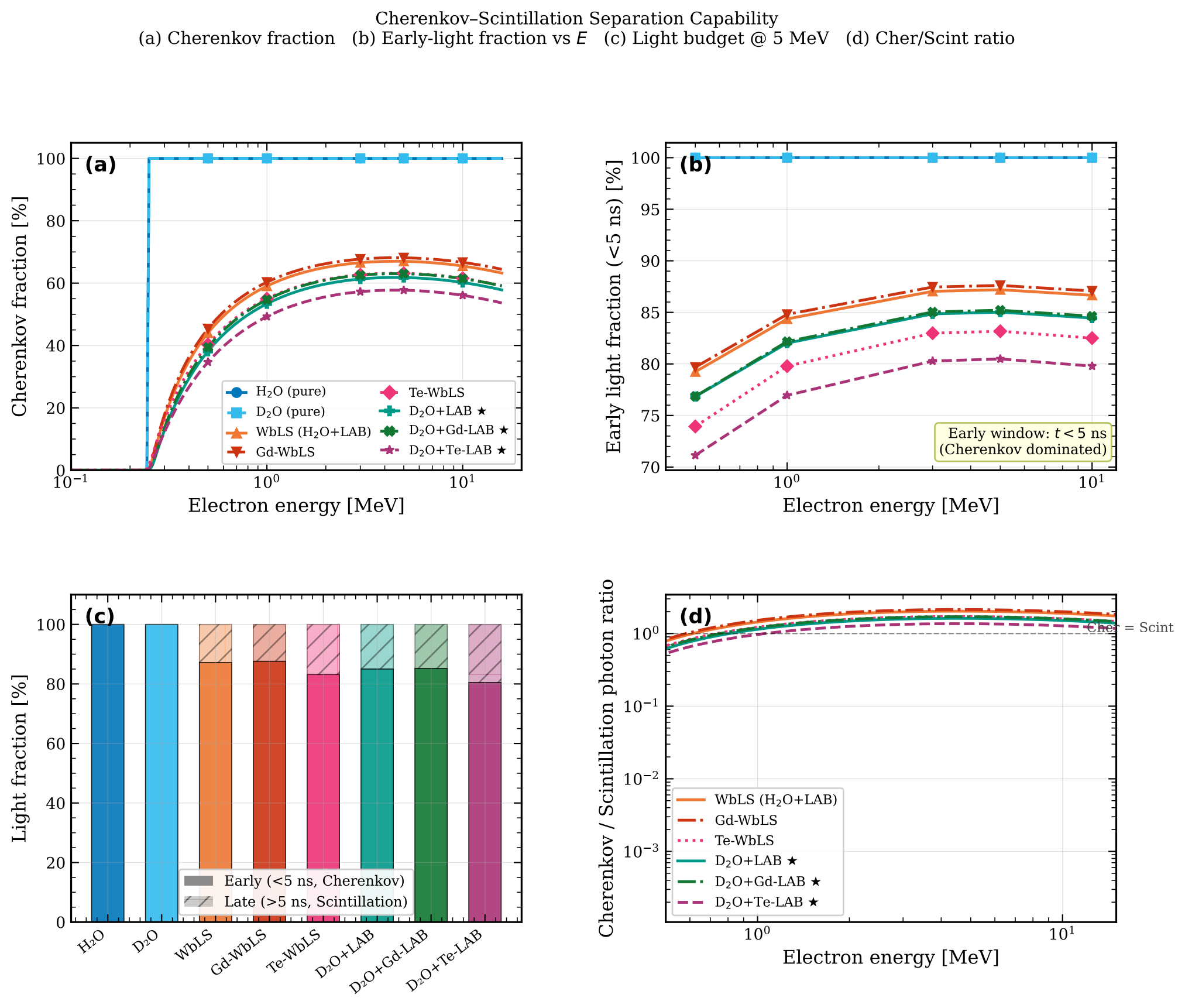
Cherenkov–Scintillation Separation Capability
(a) Cherenkov fraction (b) Early-light fraction vs E (c) Light budget @ 5 MeV (d) Cher/Scint ratio
(a)
Cherenkov fraction [%]
Electron energy [MeV]
H2O (pure)
D2O (pure)
WbLS (H2O+LAB)
Gd-WbLS
Te-WbLS
D2O+LAB ★
D2O+Gd-LAB ★
D2O+Te-LAB ★
(b)
Early light fraction (<5 ns) [%]
Electron energy [MeV]
Early window: t < 5 ns
(Cherenkov dominated)
(c)
Light fraction [%]
Early (<5 ns, Cherenkov)
Late (>5 ns, Scintillation)
H2O
D2O
WbLS
Gd-WbLS
Te-WbLS
D2O+LAB
D2O+Gd-LAB
D2O+Te-LAB
(d)
Cherenkov / Scintillation photon ratio
Cher = Scint
WbLS (H2O+LAB)
Gd-WbLS
Te-WbLS
D2O+LAB ★
D2O+Gd-LAB ★
D2O+Te-LAB ★
Electron energy [MeV]